

Harsath Kaliappa Gounder Thangavel

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Education

York University

Bachelor of Science in Computer Science (Honours)

January 2020 – December 2024

Toronto, Canada

- Data structures
- Database systems
- Computer networking
- Compiler design
- Computer architecture
- Operating systems
- Discrete mathematics
- Linguistics

Work Experience

Privex

Software Engineer

October 2023 – January 2024

London, United Kingdom

- Developed and deployed an AI customer service agent for Privex, a hosting company, using a self-hosted, private Large Language Model (LLM) and Retrieval Augmented Generation (RAG) with LangChain.
- Implemented REST APIs in Flask and integrated with Discord API to serve customers on Privex's Discord server.
- Utilized vector databases and MySQL for training data management, allowing dynamic updates from Discord by Privex staff.

CogniPet

Machine Learning Engineer

April 2019 – January 2020

Zürich, Switzerland

- Trained and deployed a deep learning model for recognizing 120 breeds of dogs using the Inception V3 neural network with 93% accuracy.
- Implemented a REST API to run model inference and integrated it with CogniPet's Android and iOS platforms using Amazon Web Services (AWS).

Projects

Jix Interpreter | *Interpreter for Jix programming language*

- Implemented an interpreter from scratch in C for a dynamically typed programming language.
- Features include support for dynamic types (int, bool, and string), arrays, functions, nested blocks, recursion, built-in functions, if-else, return, and break statements, as well as while and for loops.

Cap-Containers | *C header-only containers library*

- Highly-generic, self-contained, header-only containers written in C.
- Implements sets, maps, vectors, hash-tables (open and closed addressing), queues (circular, dynamic, fixed, and priority queues), stack, lists, and arena memory allocator.
- Includes thread-safe variants with cross-platform support (Windows and UNIX).
- Tested using CI/CD pipelines and extensively documented with Doxygen.

Blueth | *Modern C++17 Linux networking framework*

- Networking framework written in modern C++17 for Linux and provides a set of common client-server abstractions for developing high-performance, concurrent, and event-driven networking services in a modular and composable manner.
- Features include event-loop abstraction, HTTP parsers, proxies, thread-pool executors, caches, common codecs, SIMD wrappers, and networking utilities.

Poc HTTP Parser | *C header-only high-performance HTTP parsers*

- SIMD vectorized (SSE and AVX2), RFC-compliant, HTTP 1.x parsers and state-machines.
- Tested against common vulnerabilities such as HTTP request smuggling. Benchmarked using Google's microbenchmark library and compared its performance against Fastly's parsers.

Volunteer Experience

Wikipedia

Editor

July 2021 – Present

- Contributed over 20,000 edits to Wikipedia, primarily focusing on topics within computer science.
- Elevated 4 data structure and computer science articles to Good Article status.
- Collaborated with other editors to maintain the integrity and quality of the encyclopedia.

Technical Skills

Languages: Python, C++, C, Java, SQL, JavaScript, PHP, $\text{\LaTeX}$, HTML/CSS, Bash, Perl, Golang
Technologies: Linux/Unix, MySQL, Neo4j, Postfix, Dovecot, Continuous integration (CI/CD)
Developer tools: Git, Docker, LXC/LXD, CMake, Vim, GDB, Amazon Web Services